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PROCEDURAL FAIRNESS AND INTERFIRM COOPERATION IN STRATEGIC ALLIANCES

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This study applies the justice theory to address interpartner cooperation in strategic alliances. It emphasizes how procedural fairness as perceived by boundary spanners in these alliances influences cooperation outcomes. We theorize that procedural fairness improves cooperation results through enhancing relational value and curtailing relational risk in an environment characterized by both economic and social exchange. Our path analysis suggests that procedural fairness has a direct effect on operational outcome, but an indirect effect on financial outcome via increased trust driven by fairness. Procedural fairness contributes more to performance outcomes when strategic alliances are equity joint ventures than if they are contractual agreements. Theoretical and managerial implications arising from the findings are highlighted.

INTRODUCTION

Justice (fairness) is a foundation for all types of economic transactions, especially for strategic alliances that face a variety of internal and external uncertainties. Procedural justice, which concerns the extent to which the dynamics of the decision process are judged to be fair (Lind and Tyler, 1988) and the ways in which the decision-making process influences the quality of exchange relationships (Greenberg, 1986), is particularly instrumental in guiding alliance parties' behavior and contribution to interparty exchanges, and is often used as the referent to determine each party's trustworthiness and commitment in an uncertain environment (Johnson, Korsgaard, and Sapienza, 2002). Because alliance parties do not have adequate information regarding the trustworthiness of other parties nor the assurance regarding the final gains from which they share, they will refer to procedural justice to decide the extent to which they commit to joint activities. Because alliance payoff is very difficult to predict given relational risks and uncertainties (Das and Teng, 1998; Parkhe, 1993; Reuer and Koza, 2000), procedural justice can serve as one of the cornerstones that foster interfirm cooperation on the long-term basis.

Few studies, however, have addressed procedural justice in the context of strategic alliances, particularly in terms of its path in relation to cooperation outcomes in a cross-cultural setting. To partly fill this gap, this study investigates the direct and indirect links between procedural justice and alliance performances. This allows us to diagnose the channels through which justice impacts these outcomes. To this end, we propose that procedural justice has a direct-path effect on operational outcome but an indirect-path effect on financial

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outcome through heightened trust (interorganizational and interpersonal) driven by procedural justice. This study also assesses a possible moderating effect of cultural distance and governance form (equity joint venture or contractual agreement), the two main traits that often describe strategic alliances in a cross-cultural setting, on the justice–cooperation link. From a procedural justice perspective, the two traits are relevant because procedural fairness does not necessarily hold constant in relation to its outcomes, but rather is conditioned upon how an alliance is structured and how partners interact with each other (Johnson, 1997; McFarlin and Sweeney, 1992). While governance form determines alliance structuring, cultural distance affects interactions between alliance partners. Assessing these mediating and moderating effects can help executives better understand in what ways justice stimulates performance and under what circumstances justice becomes even more important to achieving economic payoffs.

Secondarily, we want to show that justice theory can enrich alliance theory in explaining the *process* of interfirm cooperation, an area that the current alliance theory still cannot sufficiently elucidate, as commented by Ring and Van de Ven (1994) and Dyer and Singh (1998). This enrichment abounds. For instance, the relational model in justice theory (Folger and Konovsky, 1989; Lind and Tyler, 1988) explicates the role of an alliance's boundary spanners in influencing the outcomes of strategic decisions and governance processes through procedural justice. Justice theory also treats procedural fairness as an important source of trust and commitment (Brockner, 2002; Konovsky and Pugh, 1994), which offers new insights into interparty cooperation. Moreover, group or social identity-based norms, such as fairness, respect, altruism, and courtesy, as stated in procedural justice theory (Allen and Meyer, 1990; Lind and Tyler, 1988), are critical to interparty attachment, conflict management, and knowledge sharing during alliance evolution.

THEORY AND HYPOTHESES

Defining procedural justice in alliance

In a general setting, procedural justice refers to individuals' perceptions about the fairness of formal procedures governing decisions. Its basic premise is that fair treatment determines individuals' reactions to decisions and is thus central to their behavior (Lind and Tyler, 1988). It emphasizes structural elements such as process control and voicing opportunity as well as interactional elements such as interpersonal treatment and work relationships as major determinants of fairness perceptions (Greenberg, 1987). Justice research has extended from how fairness influences lower-order attitudes (e.g., self-esteem, social identity, and team spirit of employees or subordinates) concerning a particular decision outcome, to how it affects higher-order, long-term attitudes such as commitment, trust, and social harmony toward groups, subunits, and institutions (Konovsky, 2000).

Financial and operational performance implications of procedural justice in strategic alliances belong to this higher-order group. This study defines procedural justice in the alliance context as the extent to which an alliance's strategic decision-making process and procedures are impartial and fair as perceived by this alliance's boundary spanners. While its essence accords with the general definition, justice in an alliance setting is distinct in several ways. First, justice perception arises from boundary spanners from both parties (for simplicity of discussion we assume only two parties in each alliance), rather than from a single party. A single party perception is unlikely to represent unbiased, objective justice. As the two parties often have varying private intents and asymmetrical dependence on alliance operations (Dyer and Singh, 1998; Inkpen and Beamish, 1997; Parkhe, 1993), it is very possible that one party manager's view of perceived fairness is different from the other party manager's view.

Second, consistent with the definition of fairness in a general context (e.g., see Greenberg, 1986; Lind and Tyler, 1988), we define 'fair' in the alliance context such that the procedures and criteria used in an alliance's strategic decisions are unbiased, representative, transparent, correctable, and ethical, as well as consistent with contractual codifications. To maintain this fairness, the parties need to establish two-way communication, mutual respect, and courtesy in the process of making and implementing strategic decisions. Lind and Tyler (1988) and Folger and Konovsky (1989) conclude that these criteria have a strong effect on individuals' attitudes about institutions and commitments to organizations. With these criteria in place, a positive perception of procedural justice can improve

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commitment through more active participation in the joint decision making and through better compliance with alliance rules and sharpened membership unity (Allen and Meyer, 1990). It is worth noting that this study joins some other studies that combine procedural and interactional justice into a single construct. In an interfirm alliance setting, interactional justice mainly involves the two-way respect or communication between organizations, which is the critical social aspect of the process or procedure in the course of interfirm exchange. It is often difficult to separate this social aspect of the procedure from the structural aspect of the procedure (Levinthal and Fichman, 1988).

Third, boundary spanners are the 'judges' of fairness in these areas; they transform perceptual justice into recommended actions that parent firms are likely to take. Increased fairness as portrayed by these spanners will curb each party's impetus for private control. Although the spirit of justice lies in the fairness of the decision process and procedure (Sapienza and Korsgaard, 1996), it is such boundary spanners who observe, conduct, and evaluate the decision process and procedures (Luo, 2001). The normative rational model of strategic decision making suggests that senior managers, such as these boundary spanners, in an alliance work in teams to make such decisions because the complexity and ambiguity of the issues with which they must grapple can overwhelm the capacities of any one individual (Hitt and Tyler, 1991). Their decision-making effectiveness depends in large part upon boundary spanners' cooperation in sharing information and in fully airing differences in assumptions and interpretations (Korsgaard, Schweiger, and Sapienza, 1995).

Finally, the process and procedures that must be viewed as fair involve important areas (decisions and their execution) such as alliance structuring, management, and interpartner sharing. In other words, fairness present in one area but not in other areas is deficient to interpartner cooperation. In the area of alliance structuring, for instance, justice about procedures in organizing the board of directors, appointing key personnel, and dividing managerial control between parties should be established in the way not only congruent with each party's commitment and contractual liabilities, but also consistent with fairness norms such as transparent communications and unbiased treatments between parties. In the area of alliance management, decisions must be jointly made by both parties using the fairness criteria such as open communications, mutual respect, and unbiased representation. In the area of interpartner sharing (knowledge, experience, and resources), each party's gains and contributions must comply with contractual descriptions—actual commitment each party made—as well as the major criteria for sharing justice, such as ethical, transparent, and reciprocal.

Procedural justice theory in alliance

In our expanded view, which integrates justice theory with alliance theory, procedural fairness is seen as a critical stimulus for improving cooperation outcomes through increased interpartner cooperation or decreased opportunism and conflicts. Fairness is especially central to socially embedded long-term economic exchanges. Justice theorists hold that fair procedures, rather than expected favorable payoffs, determine a party or an individual's behavior and commitment (Lind and Tyler, 1988; Konovsky and Organ, 1996). Because alliance payoff is very difficult to anticipate due to external and internal uncertainties, procedural fairness becomes a backbone sustaining interparty cooperation (Luo, 2005). External uncertainties often facing alliances include market changes, governmental interference, and regulatory barriers (Deeds and Hill, 1999), while internal uncertainties are frequently derived from cultural friction, incompatible objectives, asymmetric commitment, and inherent difficulty in governing and managing interorganizational relationships (Tallman and Shenkar, 1994). Fair treatment creates cooperative value by removing fears of exploitation and by demonstrating respect for the rights and dignity of the other party (Folger and Konovsky, 1989), which triggers further positive reciprocation, resulting in more stabilized relationships as well as superior citizenship behavior (Brockner and Siegel, 1995; Konovsky and Pugh, 1994; Tyler, 1989). This view enriches alliance research by offering an additional source of relational value. The relational model in justice theory further states that procedural fairness stimulates neutrality (an individual or party is treated without bias) and standing (an individual or party is treated with politeness and courtesy), which additionally enhance cooperation in repeated long-term exchanges (Konovsky, 2000; Korsgaard *et al*., 1995).

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Justice is also an important factor in reducing risks in interpartner exchange. When one party's private interest is better aligned with joint gains from the alliance via heightened procedural fairness, this party will assume greater importance to collective control because formalized procedures under joint control are neutral toward both parties, and thus viewed as safeguards against opportunism and dysfunctions. Brockner (2002) and Folger and Konovsky (1989) document that procedural fairness nourishes a party's commitment to joint efforts, increases its belief in and acceptance of collective goals and values, and strengthens its loyalty to the organization. Moreover, favorable procedural justice increases the interpartner conformity in strategic responses to major events or market changes even though consequences of such events may be unfavorable to one party (Kim and Mauborgne, 1998; Luo, 2005). This can be seen from both identification and internalization perspectives. According to Rubin and Brown (1975), identification involves group efforts for enhancing organizational affiliation and group unity, while internalization involves group efforts toward the congruence in goals and values among different members or parties of the organization. Both identification and internalization are viewed as bases for psychological attachment to organizations (Rubin and Brown, 1975). Increased procedural justice stimulates commitment and loyalty to the joint organization and countervails the 'frustration effect' that occurs because of the divergence of group norms, values, and cultures (Lind and Tyler, 1988). Shapiro and Brett (2005) suggest that fair procedures (including one party's opportunity to voice its views before a decision is made) help exchange parties feel that their outcomes will be safeguarded from any party's opportunistic behavior. They further suggest that procedural justice, especially its voice effect, can be even more important in the cross-cultural setting, but this importance is largely unexplored. Stronger group identification helps converge responses from different partners due to enhanced integrity of alliance identity and improved harmony of individual interests. Procedural justice in the strategic decision-making process also amplifies the acceptance of new internalized norms and alleviates interparty differences in managerial philosophies and styles (Allen and Meyer, 1990; Johnson, 1997). Through this internalization, both parties may become more cooperative and forbearing, underpinning the above response conformity (Das and Teng, 1998; Doz, 1996).

Further, increased procedural justice consolidates the complementary effect between collective control and cooperation. Under superior procedural justice, collective control mechanisms such as contractual and structural specifications or formulized procedures and policies are accordingly better established in terms of fairness, accuracy, transparency, consistency, and correctability (Ball, Trevino, and Sims, 1994; Greenberg, 1986; Naumann and Bennett, 2000). This improves an alliance's institutional system and governance structure in the areas of functionality and adaptability, which in turn provides a superior guidance framework under which cooperation proceeds and evolves. Meanwhile, superior procedural justice may actually help to overcome the negative effects of unilateral control. Unlike collective control, which is designed and implemented by both parties to guide and oversee shared variables in pursuit of maximum joint payoff, unilateral control is the process by which a single party ensures the alliance is managed in a manner benefiting its private variables, which may accrue at the expense of the other party's interests or joint gains. With improved procedural justice that provides a good foundation for fair exchange and impartial procedures for joint decisions and profit sharing, a party's desire for unilateral control may be reduced. Furthermore, cooperation itself is both economically structured and socially embedded (Granovetter, 1985). Superior procedural fairness consummates this economic structure through improved exchange infrastructure and corroborates social embeddedeness through improved adaptation. Justice also reduces the need for interfirm rivalry and creates a stimulating climate for cooperation. Khanna, Gulati, and Nohria (1998) argue that interfirm competition occurs when one or both parties perceive procedural unfairness and/or distributive unfairness. Johnson *et al*. (2002) observe a positive link between procedural justice and the commitment to the alliance's evolution from exchange partners and alliance management teams.

A strategic alliance's senior management team, comprising key representatives from both parties, plays a key role in helping transform procedural justice to cooperation outcomes (Luo, 2001, 2005; Schuler, Dowling, and DeCieri, 1993). As judges of justice, these representatives transform their perceptions of fairness into actions, at least partly, by their respective parent firms. Hence, they serve as the intermediaries bridging process fairness taking place in the alliance and actual control and commitment from each party. When these representatives perceive superior procedural justice, they are likely to present, on behalf of their parent organizations, more positive responses such as commitment to the joint decision, group cohesiveness, and attachment to or trust with team counterparts from the partner firm (Korsgaard *et al*., 1995; Levinthal and Fichman, 1988). Justice theorists also suggest that perceptions of fairness are created or amplified through engagement, explanation, and clarity of expectations (Greenberg, 1986; Kim and Mauborgne, 1998; Lind and Tyler, 1988). Engagement means involving individuals in decisions that affect them by asking for their input, allowing them to refute the merits of one another's ideas and assumptions, and considering their input in the final decision. Explanation means that everyone involved and affected should understand why final decisions are made and why individuals' ideas and input may have been overridden in ultimate decisions. Increased explanation adequacy is necessary if one party wants to see another party's positive response. The specificity of the explanation's substance accounts for more variance in judgments of explanation adequacy than does the interpersonal sensitivity of the explanation's delivery (Shapiro, Buttner, and Barry, 1994). Clarity of expectations requires that before, during, and after decisions are made, managers have a firm understanding of what is expected of them and what the new rules of the game are. Procedural justice differs from participative management in that the former embraces all three criteria collectively, rather than any subset of them (Korsgaard *et al*., 1995). These three means (engagement, explanation, and clarity) proceed in both an economic and social exchange environment, and their adequacy is important to the development of fairness perceptions. An alliance's boundary spanners play a key role in implementing these means.

Procedural justice and cooperation outcomes

When procedures regarding interpartner exchange and collaboration are fair as perceived by boundary spanners, group commitments from each party and its boundary spanner to the values and goals of the alliance as well as to the implementation of business strategies set forth by both parties are likely to increase, which may in turn stimulate cooperation outcomes. Cooperation outcomes are defined as alliance performance in financial return (i.e., profitability) and operational consequence (i.e., competitive strength in labor productivity, quality control, technology development, customer service, and managerial efficiency). Together, these two measures manifest both financial and operational outcomes of cooperation, enabling us to discern a detailed link between justice and performance. The group-value model in procedural justice theory views group procedure as norms for treatment and considers procedural fairness as a key determinant of an individual party's behavior even if expected private gains for this party are not favorable (Lind and Tyler, 1988). In the presence of this norm and fairness, boundary spanners become more attached to each other, resulting in a stronger alliance management team (Johnson *et al*., 2002). Because it elevates an alliance's immunity from hazardous externalities (Levinthal and Fichman, 1988), the heightened attachment can counter the disruptions from environmental and market threats that cross-cultural strategic alliances often face (Inkpen and Beamish, 1997). Further, procedural justice improves effective communication between boundary spanners and reinforces mutual understanding between investing parties (Farh, Earley, and Lin, 1997). Better communication can enhance interparty cooperation (Gulati, 1995). In addition, procedural justice generates standards of expected behavior that not only obviate the need for, but are also superior to, pure authority relations in discouraging malfeasance (Tyler, 1989). Such standards and their generalized morality and improved institutional framework strengthen process control in production, operation, and marketing, which then enhances process-based outcomes. These effects are expected to improve an alliance's operational outcome.

Procedural justice also creates a flourishing and fruitful environment for developing relationship-specific assets, which may be in the form of personal skills, or organizational routines, assets, or technologies within an alliance (Sapienza and Korsgaard, 1996; Tyler, 1989). Khanna *et al*. (1998) suggest that such relationship-specific capital results in financial benefits associated with the continuation of the relationship. Improved teamwork resulting from escalated justice also improves financial performance through enhanced coordination, learning, and routinization,

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which collectively reduce the substantial bureaucratic costs coming from internalizing operations and promote knowledge sharing between managers (Jones, Hesterly, and Borgatti, 1997; Masterson *et al*., 2000). Superior justice also heightens the internal stability of interpartner exchange. because perceptual fairness sharpens the embeddedness of formalized or routinized procedures and policies in the particular alliance relationship (Brockner, 2002; Lado, Boyd, and Hanlon, 1997). This type of binding favors resource sharing and learning between parties at both the individual and organizational levels, which may in turn give rise to high financial returns. Because justice can attenuate interparty incongruities in the corporate culture, strategic orientation, and managerial style (Shenkar and Zeira, 1992), justice development further augments an alliance's financial returns via cost reduction and managerial efficiency. In light of the above, we envisage:

Hypothesis 1: Procedural justice as perceived by boundary spanners in strategic alliances will be positively associated with alliances' operational outcome (H1a) and financial outcome (H1b).

Identifying the path or channel through which procedural justice improves operational and financial outcomes as stated in Hypothesis 1 is an important yet unexplored issue. This path reveals whether procedural justice independently and directly influences alliance performance or indirectly does so through other variables. Put differently, one needs to know if justice has a main effect on financial or operational outcomes or depends on other variables to exert its positive effect on such outcomes. We posit that, for the following reasons, justice's positive effect on operational outcome is direct, but its positive effect on financial outcome is indirect via justice-driven trust (interpersonal and interorganizational).

First, procedural fairness directly fosters operational outcome because these procedures are an inextricable part of alliance operation and interorganizational routines and, moreover, improved procedures boost operational effectiveness and foster goal achievement (Zollo, Reuer, and Singh, 2002). Operational results such as quality improvement, labor productivity, technology development, and customer satisfaction in a cooperative alliance are a direct function of the extent to which these processes are established, monitored, and implemented (Ring and Van de Ven, 1994). Fairness of various processes and procedures developed for interparty exchange and for alliance operations will significantly affect this extent since all these processes are dyadic in nature, requiring that justice be perceived by representatives of both parties. Orton and Weick (1990) suggest that a strategic alliance is a loosely coupled system in which coupling facilitates control and produces stability while looseness facilitates flexibility and produces evolution. The resulting image is a system that is simultaneously open and closed, indeterminate and rational, spontaneous and deliberate. In this situation, operational outcome of cooperation will directly rely on the fairness of procedures employed in making various decisions in this process.

Second, procedural justice indirectly improves financial outcome via trust because this justice is more distant to financial outcome than to operational outcome and because this justice contributes to trust building, which in turn contributes to financial performance. In other words, we expect that interorganizational (between two parties) and interpersonal (between boundary spanners) trust mediate the positive link between procedural justice and financial outcome. This logic emanates mainly from the attitudinal and behavioral view in procedural justice theory. Procedural fairness is associated with positive attitudes toward a decision, such as satisfaction, agreement, and commitment, and these attitudes then propel cooperative behaviors such as harmony, attachment, and trust, which shape the ultimate performance consequences for organizations (Kim and Mauborgne, 1998; Korsgaard *et al*., 1995; Masterson *et al*., 2000). In this view, individuals or parties are more apt to carefully assess the procedures used to make the decision when a decision cannot meet the preferences of all individuals or parties. This is an important notion used to assess cooperative behavior in alliances given that alliance parties often have incongruent goals, conflicting preferences, asymmetric information, and tensional needs (Park and Ungson, 1997; Reuer and Koza, 2000). Trust arises when partners perceive high justice (Das and Teng, 1998). Trust is a psychological state comprising the intention to accept vulnerability based on positive expectations of the intentions or behavior of another (Rousseau *et al*., 1998). In other words, trust is the willingness to take a risk in the partnership that is expected to create a higher payoff than pursuing it alone. The risk here concerns the extent to which there is uncertainty about whether potentially significant and disappointing outcomes of decisions or partnerships will emerge. Research on alliance trust suggests trust is a multidimensional construct containing both cognitive and affective contents and both macro (interorganizational between two parties) and micro (interpersonal between boundary spanners) elements (Sheppard and Sherman, 1998; Zaheer, McEvily, and Perrone, 1998). Research on procedural justice agrees that with greater fairness individuals or parties will work more cooperatively and more diligently under a stronger mutual trust atmosphere to ensure that joint goals are met (Brockner, 2002; Konovsky and Organ, 1996; Tyler and Blader, 2000). Conversely, acrimony limits consensus, commitment, and acceptance of decisions. As noted earlier, fairness has symbolic meaning such that having input into decisions signals that individuals are valued and respected members of the group. Individuals or parties are likely to feel more attached when they sense higher justice (Lind and Tyler, 1988). Several studies on justice have confirmed that procedural fairness increases individual or party-level trust (Ball *et al.*, 1994; Korsgaard *et al.*, 1995; McFarlin and Sweeney, 1992; Mishra and Spreitzer, 1998; Welbourne, Balkin, and Gomez-Mejia, 1995). Meanwhile, studies on alliance have validated that interorganizational and interpersonal trust improves alliance performance in financial returns (Inkpen and Currall, 1997; Mohr and Spekman, 1994; Saxton, 1997; Seabright, Levinthal, and Fichman, 1992). We hence propose:

Hypothesis 2 : Procedural justice has a direct-path effect on operational outcome (H2a) but an indirect-path effect on financial outcome through heightened interorganizational and interpersonal trust driven by procedural justice (H2b).

One prominent attribute of global strategic alliances is cultural distance between investing parties, and justice judgment and its roles are affected by cultural differences (Farh *et al.*, 1997). We anticipate that the strength of the positive link between procedural justice and cooperation outcomes as stated in Hypothesis 1 may not be constant or stable when this distance varies. Groups that view each other as more rather than less similar to each other will generally feel more attracted to, and cooperative toward, each other in strategic alliances, and cultural distance is an example of such similarity–attraction effect (Salk and Shenkar, 2001). Thus, the strength of the justice–cooperation link may differ for alliances with different cultural distances. Groups that perceive each other as procedurally just will be more likely to see positive outcomes (financial and operational) particularly in the case of shorter cultural distance, because such groups are more likely to interact with each other than when groups come from distant cultures. Interaction between groups is influenced by cultural background (Farh *et al.*, 1997). Due to this factor, the same level of procedural justice may have a stronger effect on cooperation outcomes for less culturally different groups because of the enhanced role of justice in fostering teamwork productivity (Brockner and Siegel, 1995). Because actual contributions of justice in nourishing ongoing cooperation outcomes also depend on process or procedure implementation (Korsgaard *et al.*, 1995), alliance dynamics such as interparty communication, conflict resolution, policy formalization, and incentive systems are expected to interfere with the effectiveness of process or procedural implementation. Procedural justice will contribute more to outcomes when cultural distance between partners is shorter because of lessened interference from the conflict or strengthened process effectiveness.

Moreover, the role of justice in enhancing alliance performance is impacted by cultural blending and communication effectiveness, both being substantially affected by cultural distance. Improving this role requires harmonizing two different cultures without either losing its own organizational identity in an alliance (Das and Teng, 1998). When cultural distance is smaller, cultural blending and harmonizing become easier. This helps to develop common values and norms and intensify much-needed socialization and trust building to better materialize the role of procedural justice in cooperation, especially to better streamline the cooperation process. In addition, less cultural differences between cross-national partners spur open and prompt communication between them. Only if partners can sound off on their differences will

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they be able to avoid fatal conflicts (Sheppard and Sherman, 1998). Communication also fosters the process in which one party examines the other party's credibility and trustworthiness (Rousseau *et al*., 1998). Thus, the effect of justice on performance is reinforced through proper information exchange. We suggest:

Hypothesis 3: There will be a stronger positive link between procedural justice and cooperation outcomes when interpartner cultural distance is shorter.

Likewise, the role of perceptual justice in facilitating interparty exchanges may be moderated by an alliance's importance to each party and this importance is generally manifested in alliance form (whether equity or contractual). Justice is likely to contribute more to cooperation if one or both parties view the alliance as more fundamental to their private or collective payoffs. Governance form (equity based vs. non-equity based) moderates the justice–outcome relationship through the level of interparty binding. With equity hostage and thus stronger binding in an equity joint venture, two parties are likely to attach greater importance to cooperation and be more committed to joint activities than in the case of a cooperative agreement (Osborn and Baughn, 1990). Increased commitment from each party then creates a dynamic, flourishing climate in which procedural justice will play a stronger role in enhancing cooperation and adaptation in an uncertain environment (Welbourne *et al*., 1995). Procedural justice also adds more value to resource sharing between parties in equity-based alliances than in non-equity based ones. With equity binding, both parties realize that joint payoffs largely depend on the creation of financial and operational synergies, and these synergies mainly depend on the sharing and blending of resources pooled by each party. Procedural justice becomes more important in sustaining resource commitment in the longer term. A party is willing to unilaterally commit to an equity-based alliance, even when eventual consequences are uncertain, as long as procedural and process justice is present (Gulati, Khanna, and Nohria, 1994). This unilateral commitment, however, may not emerge if the party is not equity-bound. This implies that justice contributes more to cooperation by ensuring long-term risk sharing and resource sharing when parties are more institutionally bound through equity structure. We hence suggest:

Hypothesis 4: There will be a stronger positive link between procedural justice and cooperation outcomes when a strategic alliance is equity based (as opposed to non-equity based).

METHODS

Data

This study uses China as an empirical setting to test the foregoing propositions. China is appropriate for several reasons. First, China is now the world's largest foreign direct investment recipient, approximately 60 percent of which comes from equity and non-equity (contractual) based strategic alliances (*World Investment Report*, 2006). Second, justice has been observed as a critical factor affecting Chinese people's job satisfaction, productivity or efficiency, team conflict, and individual commitment to organizations (Farh *et al*., 1997; Leung *et al*., 1996). Third, global strategic alliances in China are located in a rich research setting, ripe for examination of cooperation antecedents, behaviors, and outcomes, given the complex nature of alliance projects, cultural differences between foreign and local parties, and market dynamics during structural transformation.

This study uses data collected from three major sources: (1) the first survey, conducted in May–August 2000, targeting alliance CEOs (general managers) who responded to questions concerning alliance performance, background, trust, their own perceived justice, and the name of the chief manager representing the other party; (2) the second survey, conducted right after the first, was sent to the counterpart's chief managers whose names were offered in the first survey in order to calculate the convergent score of justice as perceived by boundary spanners from both parties; and (3) archival data were used to measure culture distance, governance form, and most control variables. Developed in both English and Chinese, the survey questionnaire was initially cross-checked by four management professors in China and further pre-tested for instrument validity with 18 alliance managers. Based on the feedback from these sources, we reworded questions or terms that were considered ambiguous.

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Sample alliances were identified from the CD-Rom Directory of Foreign-Invested Enterprises compiled by the Ministry of Commerce, China. We limited the sample to those alliances with two partners since the multiparty case (three or more) complicates the measurement of most constructs used and differs in cooperation logic and behavior from the two-party case. Our sample was further limited to alliances in manufacturing sectors (service companies do not share most of the elements of the operational outcome construct we proposed), to those in Jiangsu, Zhejiang, Shangdong, Shanghai, and Anhui provinces (major regions luring foreign direct investment for many years), and to those established for at least 3 years (most survey questions referred to the past 3-year period). With these criteria, we selected a random sample of 650 alliances.

With one round reminder, we collected 224 useful questionnaires out of 650 first surveys, representing a 34.46 percent response rate. With two round reminders, we then received 168 useful questionnaires for our second survey from 224 collected first survey samples, representing a 75 percent response rate with respect to the second survey distribution and 25.85 percent with respect to the original survey distribution. We measured justice based on the convergent score from the matched 168 sample alliances, which partly reduces the common method variance bias, and measured other survey constructs using the first-survey response. Foreign parties of these sample firms originate from Europe, the United States, Japan, and other Asian countries. Among the sample, 113 are equity joint ventures and 55 are non-equity based cooperative alliances. They participate in various sectors, from high-growth industries such as PCs, telecom, pharmaceuticals, medical equipment, and transportation equipment to low-growth ones such as plastic products and organic chemical products. These alliances invest on average U.S. $8.47 million and have an average of 577 employees.

We checked the nonresponse bias based on the information from the aforementioned CD-Rom Directory. The mean differences along major firm attributes such as total investment, foreign equity, duration, registered capital, and age between responding and nonresponding firms were contrasted using the t-test. The results showed that all t statistics were nonsignificant. We also semi-structurally interviewed 15 first-survey respondents and nine second-survey respondents to check their response accuracy (to ensure confidentiality, they were identified by code numbers stamped on each questionnaire). The results (Guttman split-half $R > 0.88$) displayed a high consistency between their reports during the interview and the answers in their survey.

Measurement

Although empirical research on procedural justice is extensive, we could find no existing scales that adequately assessed procedural justice in a manner directly relevant to interparty cooperation in the context of cross-cultural alliances. Building on theoretical dimensions found in both alliance literature (e.g., Das and Teng, 1998; Johnson *et al.*, 2002; Zaheer *et al.*, 1998) and justice literature (e.g., Kim and Mauborgne, 1998; McFarlin and Sweeney, 1992; Naumann and Bennett, 2000; Welbourne *et al.*, 1995), we developed 11 items that addressed specific aspects of procedural justice concerning interparty exchange and cooperation and strategic decision making. We first averaged the 11 items from the two surveys into a single score of procedural justice, respectively ($\alpha = 0.89$ for the first survey and $\alpha = 0.93$ for the second) and then averaged these two scores from two boundary spanners into a single index, which was further weighted by James, Demaree, and Wolf's (1984) interobserver convergence, r_{wg}, to measure the alliance's procedural justice as perceived by boundary spanners from both parties (average agreement level between first and second respondents: 0.83). In our conceptualization, fairness is considered higher if boundary spanners maintain higher agreement on justice. Variance is discounted in the index score because a difference itself in justice perception indicates attitudinal bias, and bias suppression is one of the major aims underlying procedural justice (Greenberg, 1987). Communality estimates range from 0.82 to 0.95 in the first survey and from 0.87 to 0.93 in the second survey, which validates the appropriateness of each question item. Appendix 1 describes in detail the major survey constructs.

Cooperation outcomes consist of financial and operational performances. Financial outcome was defined as return on investment (%), obtained from

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the first survey. Operational outcome was measured by the average of CEO responses (first survey) to the levels of labor productivity, manufacturing/quality control, technology development, customer service, and management efficiency, relative to major competitors in the industry. These five areas together constitute major processes in manufacturing firms and the comparison with rivals helps partial out industrial variations. To check the reliability of process response via triangulation, we asked 34 respondents in the second survey to estimate their operational outcome. The results (intraclass $R > 0.94$) showed a high consistency between boundary spanners in these operational outcome measures.

We followed recent studies on alliance trust (Inkpen and Currall, 1997; Zaheer *et al.*, 1998) to measure interpersonal and interorganizational trust. Internal consistency ($\alpha = 0.85$ and 0.80, respectively) and item appropriateness (communality $\geq$0.78 and 0.83, respectively) confirmed the reliability and validity of these two constructs. Appendix 2 also reports factor loadings of interpersonal and interorganizational trust items in a factor analysis that includes procedural justice items. Cultural distance and governance form, the two proposed moderators, were measured using archival information. Cultural distance was measured by the information from House *et al.*'s (2004) GLOBE project, which surveyed 61 nations' cultures in terms of nine dimensions (performance orientation, future orientation, assertiveness, power distance, humane orientation, institutional collectivism, ingroup collectivism, uncertainty avoidance, and gender egalitarianism). The distance measure was defined as $1/9\Sigma[(I_{ia} - I_{ic})^2/V_i]$, where V_i is the variance of cultural dimension i and I_{ia} and I_{ic} are the scores of cultural dimension i for countries a and China, respectively). The information about governance form (1 if equity-based; 0 otherwise) and country of origin was obtained from the aforementioned Directory. Finally, since cooperation outcomes are influenced by other variables as well, we controlled for some of these variables in the tests. These include (1) size (an alliance's investment scale in U.S. \$ million); (2) alliance age (the number of years present in China); (3) industry (the alliance industry's compound sales growth rate during 1997–99, computed based on the *China Statistical Yearbook*, 1998–2000 editions); (4) location (the alliance region's gross domestic product compound growth rate during 1997–99 using information from the same yearbooks); (5) market orientation (1 if export-oriented; 0 otherwise); (6) Chinese partner ownership (1 if state-owned; 0 otherwise), and (7) ethnic difference (1 if the two boundary spanners come from different ethnic backgrounds; 0 otherwise—both Chinese). The information on partner ownership, market orientation, and ethnical difference was acquired from the survey and all other information was acquired from the Directory.

RESULTS

Table 1 shows descriptive statistics and Pearson correlation coefficients between all the variables included in this study. According to this matrix, procedural justice, interparty trust, and cooperation outcomes are interrelated with one another. To uncover the link between procedural justice and alliance performance (Hypothesis 1), we performed a structural equation modeling (SEM) analysis that allows for both an omnibus assessment of the model fit and individual tests of specific hypotheses. As displayed in Table 2, the three fit indices (GFI, NFI, and CFI) in direct, indirect, and saturated models all exhibit these models' good fit to the data (GFI $\geq$0.91; NFI $\geq$0.84; CFI $\geq$0.90); and, a more complex model (saturated) provides a better fit than the direct or indirect models. The direct model (first column) was tested to verify Hypothesis 1 (the link between justice and performance). Its result suggests that procedural justice has a significant and positive effect on financial outcome ($r = 0.67$, $p < 0.05$) and operational outcome ($r = 1.03$, $p < 0.01$). Our regression analysis (Models 2 and 5 in Table 3) also presents significant links after controlling for other related variables. These results support H1a and H1b.

Our second hypothesis explores the specific path or channel through which justice influences operational outcome (H2a) and financial outcome (H2b). We questioned whether such influences proceed or are mediated through interpersonal and interorganizational trust. We compared nested models to assess the trust's mediation effects simultaneously as proposed. As Table 2 indicates (third column), the saturated model provides a fit significantly superior to both the direct model ($\chi^2{}_{\text{diff}} =$

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Table 1. Descriptive statistics and Pearson correlation matrix

Variables	Mean	S.D.	1	2	3	4	5	6	7	9	9	10	11	12	13
1 Procedural justice	5.14	1.25													
2 Interpersonal trust	5.66	1.38	0.24***												
3 Interorganizational trust	5.42	1.29	0.17*	0.19*											
4 Financial outcome	0.15	0.11	0.18*	0.20**	0.15*										
5 Operational outcome	3.22	1.09	0.27***	0.13	0.16*	0.31***									
6 Cultural distance	1.93	1.07	0.06	−0.16*	−0.03	0.08	0.13								
7 Governance form (equity)	0.67	0.47	0.09	0.09	0.12	0.05	0.10	−0.07							
8 Size	8.47	3.71	−0.05	−0.07	−0.11	−0.13	0.04	0.11	0.21**						
9 Age	9.16	1.84	0.12	0.25***	0.18*	0.20**	0.10	0.04	0.08	0.14					
10 Location	23.9	5.03	0.07	−0.04	0.03	0.12	0.09	−0.05	−0.09	0.10	0.07				
11 Industry growth	25.8	9.44	0.10	0.16*	0.07	0.22**	0.13	−0.06	0.16*	0.17*	−0.03	0.05			
12 Market orientation (export)	0.28	0.45	0.18*	0.08	0.02	0.05	0.14	0.05	−0.02	−0.15	0.03	0.11	−0.20**		
13 Local ownership (state)	0.61	0.49	−0.06	0.05	−0.16*	0.01	0.06	0.13	0.07	0.23**	0.18*	−0.02	0.04	0.07	
14 Ethnic difference	0.53	0.50	−0.15*	−0.21**	−0.04	0.09	0.13	0.27***	−0.03	0.11	−0.08	0.02	0.05	0.17*	−0.05

$N = 168$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

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Table 2. Direct and indirect effects of procedural justice: unstandardized path coefficients[a]

Variables	Direct model	Indirect model	Saturated model
Direct effect of procedural justice			
→ Financial outcome	0.67*		0.17
→ Operational outcome	1.03**		1.02**
→ Interpersonal trust	0.77*	0.72*	0.74*
→ Interorganizational trust	0.44†	0.44†	0.44†
Direct effect of interpersonal trust			
→ Financial outcome		0.51*	0.47*
→ Operational outcome		0.09	0.09
Direct effect of interorganizational trust			
→ Financial outcome		0.68*	0.69*
→ Operational outcome		0.79**	0.79**
Indirect effect of justice via interpersonal trust[b]			
Financial outcome			0.26*
Operational outcome			0.05
Indirect effect of justice via interorganizational trust			
Financial outcome			0.36*
Operational outcome			0.22
Model estimates			
Goodness-of-fit index (GFI)	0.91	0.93	0.99
Bentler and Bonett's normed fit index (NFI)	0.84	0.88	0.96
Comparative fit index (CFI)	0.90	0.91	0.97
χ^2_{diff} in nested model test I (vs. direct model)			58.22**
χ^2_{diff} in nested model test II (vs. indirect model)			31.05*

[a] All control variables were included in each model. Since these variables were not of substantive interest, their path effects are not listed in this table. Table 3 reports their regression effects on performance.
[b] The indirect path coefficient is determined by $\beta_{\text{ind}} = \beta_1 \times \beta_2$, where β_1 is a path coefficient for the effect of the independent variable on the mediator and β_2 is a path coefficient for the effect of the mediator on the dependent variable. Its significance (t-distributed) is determined by this coefficient over its new standard error $[(\beta_1 S_2)^2 + (\beta_2 S_1)^2 + (S_1 S_2)^2]^{1/2}$.
† $p < 0.10$; * $p < 0.05$; ** $p < 0.01$

58.22, $p < 0.01$) and the indirect model ($\chi^2_{\text{diff}} = 31.05$, $p < 0.05$), suggesting that a partial mediation model (i.e., both direct and indirect effects of predictors) provides the best fit of the data. To demonstrate mediation for specific relationships, we investigated the three conditions necessary for mediation as stated by Baron and Kenny (1986). First, the predictor (justice) must be related to the mediator (trust). The first column of Table 2 shows that justice is significantly linked with interpersonal trust ($r = 0.77$, $p < 0.05$) and interorganizational trust ($r = 0.44$, $p < 0.10$). Second, the mediator (trust) must be related to the dependent variables (financial and operational outcomes). The second column of Table 2 shows that interorganizational trust is significantly related to both financial ($r = 0.68$, $p < 0.05$) and operational outcome ($r = 0.79$, $p < 0.01$) while interpersonal trust is significantly related to financial outcome only ($r = 0.51$, $p < 0.05$). Third, the previously significant relationship between the predictor variable (justice) and dependent variables (outcomes) should be eliminated or substantially reduced when the mediator (trust) is accounted for. The direct effect of justice on financial outcome, significant in the direct model ($r = 0.67$), now disappears in the saturated model ($r = 0.17$). However, the direct effect of justice on operational outcome remains the same after the mediators are included. All these results suggest that trust, both interpersonal and interorganizational, mediates the influence of justice on *financial* outcome but not on *operational outcome*. To further validate these results, we examined the significant level (t-distribution) of indirect path coefficients (see the saturated model). The test reports that the indirect effects on financial outcome via interpersonal trust ($r = 0.26$, $p < 0.05$) and interorganizational trust ($r = 0.36$, $p < 0.05$) are both significant. Such indirect effects on operational outcome are not. Together, these results demonstrate that procedural justice has a direct-path effect on

Table 3. Cultural distance and governance form as moderators: hierarchical moderated regression[a]

Variables	Financial outcome			Operational outcome		
	1	2	3	4	5	6
Intercept	5.03*	7.34*	6.68**	1.69***	1.45*	2.65***
	(2.34)	(4.07)	(1.88)	(0.33)	(0.63)	(0.72)
Main effects						
Procedural justice (X1)	0.16*	0.16*	0.09	0.25**	0.24**	0.25*
	(0.07)	(0.07)	(0.08)	(0.09)	(0.08)	(0.14)
Cultural distance (M1)		0.07	0.07†		0.07	0.06
		(0.04)	(0.04)		(0.05)	(0.06)
Governance form (equity) (M2)		0.08	0.12		0.05	0.14
		(0.10)	(0.11)		(0.10)	(0.12)
X1 × M1			0.01			0.03
			(0.04)			(0.05)
X1 × M2			0.28*			0.38**
			(0.13)			(0.15)
Control variables						
Interpersonal trust	0.14†	0.14†	0.14†	0.04	0.04	0.03
	(0.08)	(0.08)	(0.08)	(0.06)	(0.06)	(0.06)
Interorganizational trust	0.20**	0.18*	0.18*	0.12*	0.12*	0.13*
	(0.08)	(0.08)	(0.09)	(0.05)	(0.05)	(0.06)
Size	−0.05	−0.05	−0.05	0.09	0.09	0.10
	(0.06)	(0.05)	(0.05)	(0.11)	(0.11)	(0.11)
Age	0.28**	0.24**	0.23*	0.17	0.15	0.14
	(0.10)	(0.10)	(0.10)	(0.13)	(0.13)	(0.13)
Location	−1.30	−1.19	−1.00	−0.33	−0.48	−0.43
	(1.41)	(1.35)	(1.27)	(1.11)	(1.09)	(1.10)
Industry growth	0.86**	0.76**	0.74**	0.20	0.18	0.20
	(0.33)	(0.31)	(0.31)	(0.28)	(0.28)	(0.29)
Market orientation (export)	0.08	0.08	0.07	0.10	0.07	0.07
	(0.14)	(0.14)	(0.13)	(0.13)	(0.12)	(0.12)
Local ownership (state)	0.09	0.10	0.11	0.08	0.07	0.06
	(0.12)	(0.12)	(0.12)	(0.10)	(0.10)	(0.11)
Ethnic difference	0.06	0.07	0.06	0.10	0.09	0.09
	(0.07)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)
Model F	10.10***	11.79***	21.90***	7.79***	9.76***	15.13***
Adjusted R^2	0.37	0.39	0.44	0.31	0.32	0.39
Δadjusted R^2		0.02	0.05		0.01	0.07
Hierarchical F		1.67	6.35**		1.06	8.24***

[a] The entries are unstandardized βs with standard errors in parentheses. The interaction terms are centered as recommended by Aiken and West (1991).
†$p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

operational outcome but an indirect-path effect on financial outcome via trust. H2a and H2b are hence accepted.

We conducted a hierarchical moderated regression to verify Hypotheses 3 and 4 (moderating effect of cultural distance and governance form). The variance inflation factor values of all independent and control variables in Models 2 and 5 in Table 3 range from 1.06–2.88, suggesting the absence of the multicollinearity before including interaction terms. To remove the inherent multicollinearity between predictor variables and interaction terms that include these predictor variables, we centered the two moderating variables around the mean as suggested by Aiken and West (1991). The regression results suggest that cultural distance (M1) and governance form (M2) do not have a main effect on either financial outcome (Model 2) or operational outcome (Model 5). As shown in Models 3 and 6, the interaction between justice and governance form (X1 × M2) is significant and positive in relation to financial outcome ($\beta = 0.28$, $p < 0.05$) and operational outcome ($\beta = 0.38$, $p < 0.01$). The

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interaction between justice and culture (X1 × M1) is significantly related neither to operational nor financial outcomes. This evidence suggests that governance form does not independently affect alliance performance but does moderate the positive link between justice and performance (both financial and operational outcome) such that there will be a stronger justice–performance relationship if the alliance is equity-based. Cultural distance does not independently affect performance nor moderate the effect of justice on performance.

To further validate the above moderating effects, we plotted the interactions by inserting high and low values equal to one standard deviation above and below the mean for each moderator variable, respectively. Panels I and III of Figure 1 reveal that procedural justice is associated with financial performance (I) and operational performance (III) in a similar fashion regardless of the level of cultural distance. However, the interaction in panel II is noteworthy, because it illustrates that the positive slope of the justice–financial outcome relationship is fundamentally steeper for equity-based alliances (high level) than for non-equity-based (low level). Panel IV also shows a significantly steeper slope for equity-based alliances than for non-equity based (contractual) concerning the association between procedural justice and operational performance. In all, these results suggest that governance form moderates the relationship between procedural justice and cooperation outcomes but cultural distance does not. In equity-based alliances, justice contributes more to financial and operational performances. This finding lends support to Hypothesis 4 but not to Hypothesis 3.

DISCUSSION AND CONCLUSION

This study applies the justice theory to explain how procedural fairness affects cooperation outcomes. It extends individual-focused justice research to the interfirm level and to the context of cross-cultural strategic alliances. Alliances need procedural justice when formulating, governing, and managing interorganizational entities because this justice serves as a foundation for interparty cooperation, knowledge sharing, economic exchange, and ongoing commitment. Our analysis suggests that procedural justice as perceived by boundary spanners in alliances has a positive influence on cooperation

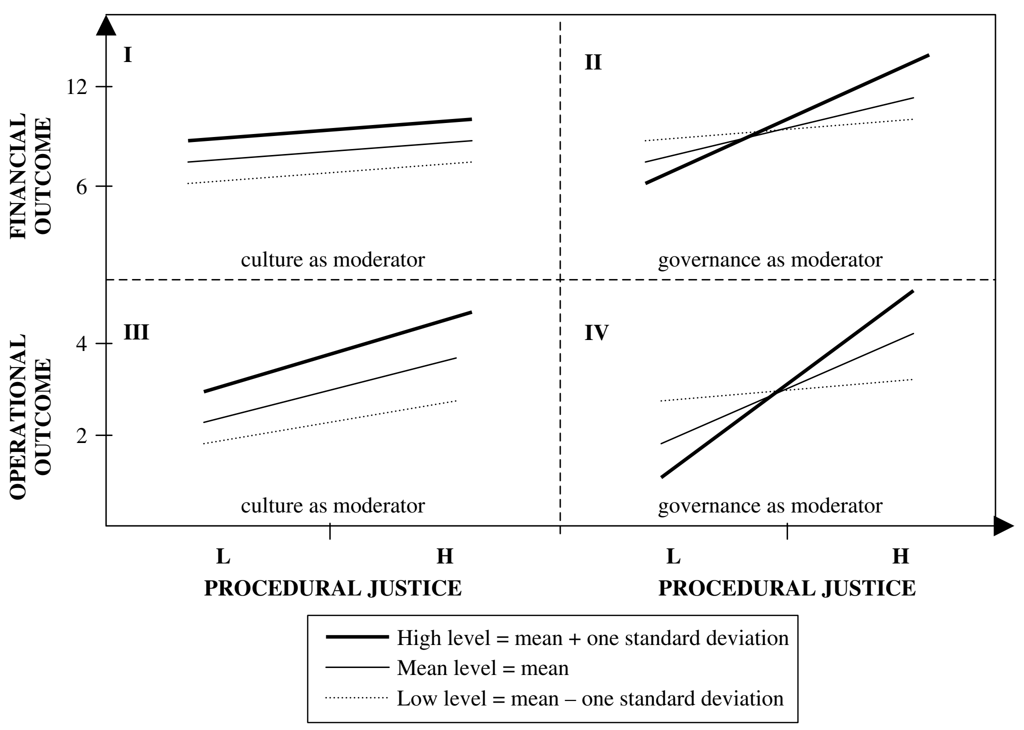

Figure 1. Plotting the interactions

outcomes in the form of both operational and financial performance. While procedural justice directly contributes to operational outcome, it indirectly improves financial outcome via heightened interpersonal (between boundary spanners) and interorganizational (between parties) trust driven by procedural justice. This path difference implies that procedural justice more directly affects operational consequence than financial consequence and shows how justice, trust, and outcome are interrelated with one another. The direct effect of justice suggests that process fairness boosts process effectiveness, *ceteris paribus*. Superior fairness improves the governance framework within an alliance, thus better guiding operational processes in production, innovation, and marketing and better enlightening organizational processes in general management and productivity enhancement. The indirect link between procedural justice and financial performance suggests that justice stimulates interpersonal and interorganizational trust, which then stimulates financial performance.

While trust mediates the relationship between justice and financial outcome, governance form is found to moderate the relationship between justice and cooperation outcomes. Justice contributes more to performance of equity-based alliances than to contractual alliances because of the 'equity hostage' effect, under which procedural fairness becomes more important to collective gains in repeated long-term economic transactions involving large sums of capital contributions from all parties. Contrary to our expectation, cultural distance is not found to intervene with the justice–performance link. We caution that cultural distance in this study focuses on national-level differences, which may not reflect corporate-level and individual-level culture differences. For firms with rich experience in the host country, their long cultural distance at the national level may be considerably undercut by their short distance at the corporate level. The finding that financial performance is an increasing function of alliance age (Table 3) corroborates this view and exhibits the importance of experience effect. Because boundary spanners (top management team) are deemed to be pivotal in transmitting perceived justice into parent actions, individual-level cultural distance can be especially important in determining the agreement of perceived justice between representatives and thus affecting concert actions by all parties. Executives from a home country culturally distant from a host country can curtail the negative effect of national cultural distance by strengthening the corporate- and individual-level adaptation to local culture and amplifying their personal trust with foreign counterparts. Future research should continue to advance this line of inquiry and explore how individual cultural traits (e.g., background and experience) influence justice perceptions, trust building, and decision-making quality and how individual-level cultural adaptations offset the deterrence of national- and corporate-level cultural differences in the process of alliance exchange. It is also intriguing to see how boundary-spanning individuals socially lubricate and alleviate national- and corporate-level cultural conflicts through acculturation, communication, adaptation, and socialization.

Despite its importance in shaping alliance evolution, procedural justice has been rarely applied to explain interfirm cooperation. In the alliance context, 'justice' is perceived by boundary spanners from both parties, rather than a single party, and is thus dyadic in nature. 'Procedure' involves an array of strategic decisions in the process of establishing, organizing, running, and managing the alliance, thus extending beyond the process of control or formalization. In our view, procedural justice influences both the process and structure of interpartner exchange, the two central concepts in alliance research. If the process itself is fairer, it may add more relational value to both parties arising from increased mutual commitment and sustained cooperation. Justice also consummates the structure by alleviating relational risks arising from unilateral opportunism and private interest pursuit. Together, as our study documents, procedural justice bolsters joint payoffs in the form of operational and financial outcomes. With this confirmed finding, alliance research may be enriched by unifying justice, process, and structure and specifically integrating justice's role in fostering cooperation with governance mechanisms (e.g., managerial control, formalization, and contract), trust building (individual and organizational), adaptive commitment, and boundary-spanning socialization. Justice itself contains both economic elements (fair procedure and process) and social elements (respect and courtesy), and is thus likely to nurture or temper the developmental process of socially embedded economic exchanges.

The major findings of this study provide several new avenues for future research on strategic

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alliances. First, we demonstrate a positive link between justice and cooperation, but how justice influences control is unknown. All alliances involve some balance between cooperation and control since partner firms are barely completely opportunistic nor totally trusting. It would be particularly interesting to see if procedural justice affects private control and joint control differently. One may expect that procedural justice can circumvent private control but support joint control. This, however, remains to be verified. Second, we observe a positive link between justice and trust, but how justice influences commitment is unknown. Justice theorists hold that procedural fairness solidifies individual commitment to the organization. The question remains as to whether fairness can augment each partner firm's commitment to the joint organization. According to alliance theory, it is commitment, rather than trust, that directly determines alliance performance, but trust fosters commitment and contribution from all parties (Das and Teng, 1998; Johnson *et al.*, 2002). Whether a positive link between trust and performance is propped up by commitment and whether commitment mediates the justice–cooperation link are two critical questions to be explored. Third, we document that the justice–performance link is moderated by governance form, but how governance frameworks or mechanisms are interrelated with procedural justice is unknown. Both the process and content of governance frameworks such as contractual stipulations and formalized policies, procedures, and rules necessitate procedural justice to improve the framework quality and, more importantly, to nourish the functioning of these frameworks. Future endeavors may decipher the path and vigor of justice's role in fortifying governance frameworks and discern the three-way interactions between governance form, governance structure, and procedural justice.

Furthermore, we report the results in an empirical setting of an emerging market (China) after controlling for location and industry factors, but how market dynamics or environmental conditions in a typical emerging market specifically influence the role of procedural justice in strategic alliances is unknown. Justice may become more imperative, but more difficult to develop, under greater market uncertainty. Many emerging markets like China have a long tradition of relying more on social or personal connections to get things done and depending less on judiciary and justice systems. It would be interesting to see the extent to which environmental conditions as well as traditional heritages of these societies actually affect the role of procedural justice in stimulating alliance development. The answers to the above questions lie in the realm of dynamics of justice, alliance, and market, for which this study's sample is not yet ready to provide illumination but future research should explore.

Finally, this study focuses on how procedural justice as perceived by alliance-level boundary spanners affects alliance performance outcomes. Although we include several attributes of multinational organizations in our analysis, we do not examine how parent firms of multinational organizations play their roles in the establishment and function of procedural justice in foreign alliances they build. Boundary spanners representing these organizations are certainly not isolated from the influence by their upper authorities at the corporate office. Their anticipation of and commitment to procedural justice needed in foreign alliances may change according to multinational organizations' corporate policy and strategic objective relating to these alliances. Thus, the role of procedural justice in improving alliance performance can be affected by multinational organizations' intent and policy. For instance, multinational organizations seeking greater national adaptation or local responsiveness may attach greater importance to the role of procedural justice in foreign alliances than those organizations pursuing higher integration and control. Future investigation that addresses these issues is expected to advance our understanding of procedural justice in a broader context that straddles not only national borders, but also multiple levels within a geographically dispersed yet globally coordinated organization.

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Appendix 1: Selected Survey Items[a]

1. *Procedural Justice* (seven-point Likert scale, from 1 = not true at all to 7 = very true)
    (1) There is a two-way communication between two parties in the process of making strategic decisions relating to alliance operations and management
    (2) I am always treated with respect and courtesy in the above process by my counterpart (top manager from the other party)
    (3) My counterpart always confers with me before taking new initiatives about the alliance
    (4) The procedures used by the alliance's board members in their decision-making process are impartial
    (5) The procedures used by two parties in negotiating, stipulating, and codifying the alliance contract are fair
    (6) The procedures used by two parties in formulating and structuring the alliance are fair
    (7) The procedures used by two parties in planning, organizing, and managing alliance activities are fair
    (8) The procedures used to govern knowledge or resource sharing between two parties are impartial and fair
    (9) The procedures of executing strategic decisions are clearly defined and performed consistently
    (10) The execution of the alliance contract is administered and monitored fairly by both parties
    (11) The implementation of strategic decisions is administered and monitored fairly by both parties
2. *Interparty Trust* (seven-point Likert scale, from 1 = not true at all to 7 = very true)
    A. Interpersonal trust
    (1) My counterpart can always be counted on to act as I expect
    (2) My counterpart is trustworthy
    (3) My counterpart and I can always find appropriate solutions through compromise when conflicts arise
    (4) In a tough time of alliance operations, my counterpart and I rely on and get help from each other
    (5) I always feel confident when my counterpart tells me he will do something
    (6) My counterpart and I always share information and experience about management and even personal life
    (7) My counterpart always shares or takes responsibilities for managerial or operational problems even if he should not be obligated for these
    (8) My counterpart and I engage in important activities even if these activities are not explicitly documented in an alliance agreement
    B. Interorganizational trust
    (1) The partner firm in our alliance can be trusted to make sensible alliance decisions
    (2) The partner firm in our alliance is quite prepared to gain advantage by deceiving our party[b]
    (3) Both parties in our alliance can rely on each other to abide by the alliance management agreement
    (4) Our party is reluctant to make resource commitment to the alliance when specifications in the alliance agreement are ambiguous[b]
    (5) Both parties in our alliance have a high level of mutual trust in various activities
    (6) The partner firm always stands by its word even when this was not in the best interest for it
    (7) The partner firm never uses opportunities that arise to profit at our expense
    (8) The partner firm is flexible when our party cannot keep a specific promise stipulated in alliance agreement due to unexpected environmental change
3. *Operational Outcomes*
    Please circle the number best estimating how your firm compares to close competitors in your industry on each of the following over the past three years (1 = lowest 20% in the industry, 2 = lower 20%, 3 = middle 20%, 4 = next 20%, 5 = top 20%):
    (1) Labor productivity
    (2) Manufacturing/quality control
    (3) Technology development
    (4) Customer service
    (5) Management efficiency

[a] The informants were asked to refer the past 3 years as the time frame to answer the above survey questions.
[b] Reverse coded item.

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Appendix 2: Factor Loadings of Justice and Trust (Varimax Rotation)

	Variable items	Factor 1	Factor 2	Factor 3
Procedural justice				
1.	Two-way communication	0.81	0.37	−0.17
2.	Respect and courtesy	0.83	0.41	−0.10
3.	Conference on decisions	0.79	0.43	−0.15
4.	Impartial board decisions	0.84	0.42	−0.09
5.	Fairness in building alliance	0.86	0.37	0.02
6.	Justice in structuring alliance	0.83	0.41	0.03
7.	Fairness in managing alliance	0.84	0.40	−0.07
8.	Knowledge sharing fairness	0.82	0.43	−0.08
9.	Strategic decision procedures	0.81	0.37	−0.16
10.	Contract execution fairness	0.83	0.44	−0.04
11.	Decision implementation	0.85	0.37	−0.05
Interpersonal trust				
1.	Counterpart accountability	0.15	0.26	0.74
2.	Counterpart trustworthy	0.16	0.31	0.67
3.	Mutual compromise	0.15	0.27	0.74
4.	Mutual support	0.12	0.30	0.71
5.	Confident on counterpart	0.10	0.28	0.68
6.	Information sharing	0.18	0.36	0.63
7.	Responsibility sharing	0.20	0.25	0.65
8.	Activity sharing	0.13	0.33	0.67
Interorganizational trust				
1.	Partner trustworthy in decision making	0.43	0.81	−0.06
2.	Partner deceiving behavior (reverse coded)	−0.12	0.94	0.02
3.	Honoring agreement	0.47	0.80	0.06
4.	Reluctance in contribution (reverse coded)	0.13	0.75	0.11
5.	Mutual understanding	0.44	0.82	0.09
6.	Standing by words	0.45	0.80	0.14
7.	Resisting opportunism	0.24	0.71	−0.43
8.	Partner flexibility	0.50	0.78	0.01
Variance		9.55	8.42	6.49